

Set 35: Solutions of acids and bases

Notes

Dilutions

When a solution is diluted, the volume increases and the concentration decreases. The total amount of solute, that is, the number of moles, remains the same. This can be expressed as:

$$\begin{array}{rcl} n_1 \text{ (before dilution)} & = & n_2 \text{ (after dilution)} \\ c_1 V_1 & = & c_2 V_2 \end{array}$$

Example

To prepare 250.0 mL of 0.200 mol L⁻¹ nitric acid, what volume of concentrated 15.6 mol L⁻¹ nitric acid would be required?

(a) Calculate the known number of moles:

$$\begin{array}{rcl} n(\text{dilute HNO}_3) & = & c(\text{HNO}_3)(V) \\ & = & (0.200)(250.0 \times 10^{-3}) \\ & = & 5.00 \times 10^{-2} \text{ mol} \end{array}$$

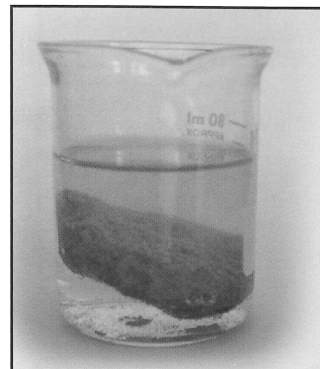
(b) Relate the unknown number of moles of concentrated HNO₃ to the known number of moles of dilute HNO₃:

$$\begin{array}{rcl} n(\text{concentrated HNO}_3) & = & n(\text{dilute HNO}_3) \\ (15.6)(V) & = & 5.00 \times 10^{-2} \\ V & = & \frac{5.00 \times 10^{-2}}{15.6} \\ V & = & 3.21 \times 10^{-3} \text{ L} \end{array}$$

Set 35: Exercises

- 25.6 g of anhydrous sodium carbonate is dissolved in 200.0 mL of water.
 - Calculate the concentration in grams per litre of this solution. Assume no volume change.
 - Calculate the concentration in mol L⁻¹.
 - A 20.0 mL sample of this solution was placed in a 150.0 mL flask and 80.0 mL of water added. Determine the concentration of sodium ions in this diluted solution.
- Calculate the concentration of
 - chloride ions in 25.0 mL, of 0.200 mol L⁻¹ hydrochloric acid solution
 - hydroxide ions in 2.20 L of 2.02×10^{-3} mol L⁻¹ barium hydroxide solution
- 10.0 g of calcium nitrate is dissolved in 220.0 mL of distilled water. Calculate the concentration in mol L⁻¹ of:
 - Ca²⁺ ions
 - NO₃⁻ ions
- Calculate the volume of 10.0 mol L⁻¹ nitric acid solution required to prepare 500.0 mL of 0.500 mol L⁻¹ solution.
- Calculate the concentration in mol L⁻¹ of hydroxide ions in a solution prepared by mixing 360.0 mL of 0.250 mol L⁻¹ sodium hydroxide solution with 675.0 mL of 1.20 mol L⁻¹ potassium hydroxide solution. Assume solution volumes are additive.
- What volume of water must be added to 150.0 mL of 1.10 mol L⁻¹ sulfuric acid solution to prepare a 0.210 mol L⁻¹ solution?

7. Battery acid is made by diluting (98.0% by mass) sulfuric acid with water. If 1.00 kg of the 98.0% acid was mixed with water to make up 3.00 L of battery acid, calculate the concentration of the resulting acid in g L^{-1} .
8. A 0.100 mol L^{-1} solution of hydrochloric acid was required to analyse some sandstone. Calculate the mass of hydrogen chloride gas required to make 300 mL of this solution.
9. Slaked lime (calcium hydroxide) can be used to reduce soil acidity and is sparingly soluble in water. Calculate the concentration in mol L^{-1} of:
 - (a) calcium ions;
 - (b) hydroxide ions;
 in a solution containing 10.0 mg of calcium hydroxide in 1.00 L of solution.
10. Where vegetables are grown hydroponically, fertilisers are applied in solution. Ammonium nitrate, potassium sulfate and calcium dihydrogenphosphate are typical compounds used.
 - (a) Write equations to illustrate how each of these compounds result from acid-base reactions.
 - (b) Calculate the mass of solute required to prepare each of the solutions listed:
 - (i) 10.0 L of 1.50 mol L^{-1} ammonium nitrate
 - (ii) 2.50 L of 2.80 mol L^{-1} potassium sulfate
 - (iii) 500 mL of 0.100 mol L^{-1} calcium dihydrogenphosphate
11. Concentrated sulfuric acid needs to be diluted before it is used to dissolve a sample of iron prior to analysis. Calculate the volume of 18.0 mol L^{-1} sulfuric acid required to prepare 800 mL of a 2.50 mol L^{-1} solution.
12. A mixture of hydrochloric acid and nitric acid is used to digest (decompose) organic material, so that meat can be analysed for lead, copper, cadmium and zinc. Calculate the concentration of hydrogen ions, in mol L^{-1} , in such a solution prepared by mixing $3.60 \times 10^2 \text{ mL}$ of a 14.3 mol L^{-1} nitric acid solution with 675 mL of 12.1 mol L^{-1} hydrochloric acid solution. Assume the solution volumes are additive.
13. Bore water with a hydrogen ion concentration of $3.60 \times 10^{-4} \text{ mol L}^{-1}$ can be relatively acidic. This can make it unsuitable for irrigation. Mixing this water with rain water reduces the concentration of hydrogen ions and hence increases the pH.
 - (a) Assuming the rain water is neutral, calculate the volume of rain water that needs to be added to $3.00 \times 10^3 \text{ L}$ of bore water to reduce the hydrogen ion concentration to $1.00 \times 10^{-6} \text{ mol L}^{-1}$.
 - (b) Is this a practical method of reducing the hydrogen ion concentration? Explain.
 - (c) Suggest other methods that could be used to achieve the same result. How does this compare with the addition of rain water?
14. A dilute sulfuric acid solution can be used to dissolve magnesite samples prior to analysis. Magnesite contains mostly magnesium carbonate. Calculate the volume of water that must be added to $1.50 \times 10^2 \text{ mL}$ of a 5.50 mol L^{-1} sulfuric acid solution to prepare a 0.500 mol L^{-1} solution for use in the analysis.



Question 8: Sandstone in hydrochloric acid



Question 10: Hydroponics (photo: Joel Malcolm)